The Effects of Productivity and Demand-Specific Factors on Plant Survival and Ownership Change in the U.S. Poultry Industry

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Abstract

In this paper we study the productivity-survival link in the U.S. poultry processing industry using the longitudinal data constructed from five Censuses of Manufactures between 1987 and 2007. First, we study the effects of physical productivity and demand-specific factors on plant survival and ownership change. Second, we analyze the determinants of the firm-level expansion. The results show that higher demand-specific factors decrease the probability of exit and increase the probability of ownership change. The effect of physical productivity on the probability of exit or ownership change is generally insignificant. Also, firms with higher demand-specific factors have higher probability to expand whereas the average firm-level physical productivity turns out to be an insignificant determinant of firm expansion.

Keyword: Productivity; Demand-specific Factors; Poultry

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THE EFFECTS OF PRODUCTIVITY AND DEMAND-SPECIFIC FACTORS ON PLANT SURVIVAL AND OWNERSHIP CHANGE IN THE U.S. POULTRY INDUSTRY

1. Introduction

The literature on firm dynamics emphasizes the role of firm specific productivity shocks in accounting for firm entry and exit. As a consequence of this idiosyncratic uncertainty, substantial amounts of resources are reallocated across firms from shrinking and exiting ones to entering and expanding ones. Empirical research in this area generally finds that plants with higher measured productivity levels tend to grow faster and are more likely to survive. The examples are found in Baily, Hulten and Campbell (1992) for the U.S. manufacturing plants; Olley and Pakes (1996) for the U.S. telecommunication equipment industry; Dwyer (1995) for the U.S. textile industry; and Baldwin (1995) for the Canadian manufacturing sector. Most of this literature relies on the so called traditional productivity measure, i.e., revenue divided by a common industry-level deflator as a measure of output. This is because the plant-level data on physical output is typically not available; only plant-level revenue is recorded. This approach may be acceptable if product quality differences are fully reflected in prices. However, this can be problematic if other factors, such as demand-specific factors which vary across producers, are embodied in price variations across producers for homogenous products.[5] In this case, traditional productivity (revenue-based productivity) confounds both the effects of physical productivity and demand-specific factors on market selection.

The problems associated with the measure of traditional productivity have been identified for quite some time. Abbott (1992) documented the extent of price dispersion within various industries and outlined possible implications for the measurement of aggregates. Klette and Griliches (1996) and Mairesse and Jaumandreu (2005) considered how intra-industry price fluctuations affect production functions and productivity estimates. Katayama, Lu and Tybout (2009) demonstrated that revenue-based output and expenditure-based input measures can lead

[5] In theoretical models, demand-specific factors are modeled as consumers' preference of one product over another; see Foster, Haltiwanger and Syverson's (2008). One potential source of demand-specific factors could be the transportation cost that could reflect demand idiosyncrasies across local markets which can give rise to some localized market power even in industries producing homogenous products (commodities). Another source of horizontal differentiation is complex, history-laden, frequently totally non-transparent, collection of relationships between producers and buyers.

to productivity measurement error and incorrect interpretations about how heterogeneous producers respond to shocks and associated welfare implications.

In this paper we study the productivity-survival link in the U.S. poultry processing industry using the longitudinal data constructed from five Censuses of Manufactures between 1987 and 2007. In the Census data we observe plant level prices and quantities which enable us to separately estimate physical productivities and plant level demand functions. The motivation to study the dynamics of the broiler processing industry comes from two observations. First, during the last two decades, the broiler industry has been very dynamic. Based on the Census data (which will be discussed in details below), the industry became more concentrated (the Herfindahl index increased from 735 to 1224 from 1992 to 2007). During the same time period, 59 out of 471 U.S. broiler processing plants closed down and 53 were acquired by other firms. Such intense closure, mergers and acquisitions activity provides a compelling reason to study the mechanisms that can explain these events. Second, the broiler processing industry is an ideal candidate to investigate the importance of demand-specific factors on industry dynamics. The industry segment that we investigate produces a reasonably homogenous product (chicken meat), yet plant-level or firm-level prices vary because of the geographic demand variation or because of the existence of historically determined special business (or personal) relationships between producers and their customers.[6]

Plant-level industry dynamics in meat packing industries have been studied before. Anderson et al. (1998) used plant-level data of the U.S. beef packing industry from 1991 to 1993 to study the effect of plant-level characteristics, market structure, and supply and demand shifters on the probability of plant exit. They found that the effects of plant and market structure characteristics are significant while the effects of supply and demand shifters are not. Muth et al. (2002) analyzed plant-level data in the U.S. red meat industry from the pre-HACCP (Hazard Analysis and Critical Control Points) period (1993 to 1996) and the implementation period (1996 to 2000). They used plant's age as a proxy for productivity and found that very young or very old plants have a higher probability of exit. Muth et al. (2003) extended their earlier work to include

[6] For example, there is anecdotal (public) evidence about special business ties between Tyson and Walmart, probably originating from the fact that both companies started and are still headquartered in the same, relatively small, state (Arkansas). It is interesting to see that an excellent assortment of various meat products sold by Sam's Club (owned by Walmart) is always supplied by Tyson. The existence of some special deals or long-term supply arrangements between those two companies would be impossible to document. References to specific companies (e.g., Tyson) in the text are based on public information.

poultry slaughter and processing-only plants in addition to red meat slaughter plants. By considering this wider set of industries, they were able to compare the effects of different industry structures on the probability of exit. Nguyen and Ollinger (2006) investigated the motives for mergers and acquisitions in the U.S. meat products industry relying on the Census longitudinal data from 1977 to 1992. Their results show that acquired meat and poultry plants were very productive before mergers and that the majority of plants significantly improved their productivity growth after the merger. Nguyen and Ollinger (2009) evaluated the impact of mergers and acquisitions on wages, employment, and plant closures using two balanced panels of all plants owned by meat and poultry firms that existed in the 1977 Census of Manufacturing and survived until 1987 and those that existed in 1982 and survived until 1992. They found that that mergers and acquisitions are positively associated with wages and employment during 1977–1987, but not during 1982–1992. Ollinger (2011) used plant-level micro-data over 1987–2002 to examine structural change in the cattle and chicken slaughter and pork processing and sausage-making industries during a period of increasing food safety oversight. The results suggest that HACCP rules may have favored large cattle slaughter plants and large chicken slaughter plants during the implementation period spanning the 1997–2002 period but there is no evidence that events favored large plants over small plants in pork processing and sausage-making.

The most closely related study to ours is Foster, Haltiwanger and Syverson's (2008) where authors observe plant-level quantities and prices so they could measure physical productivity and plant level demand-specific factors separately. By using a sample of approximately 18,000 plant-level observations in eleven homogenous products industries, they found that higher "revenue-based productivity" increases the probability of survival. After separating revenue-based productivity into "physical productivity" and the plant-level "demand-specific factors", they concluded that both physical productivity and plant level demand-specific factors increase the probability of survival, but the dominant factor determining survival are the plant level demand-specific factors. This shows that "revenue-based productivity" overstates the effects of productivity on survival.

In addition to studying the effects of physical productivity and demand-specific factors on plant survival, we also study the importance of these effects on ownership change. Most previous studies treated acquired plants as exits (e.g. Anderson et al. 1998) which could be problematic to

the extent that plants that have completely exited the industry could be fundamentally different from plants that have been acquired by other firms. Finally, given the fact that the poultry industry grew substantially during the studied period, we also analyze the determinants of the firm-level expansion. As a brief preview of the results, we were able to show that higher demand-specific factors decrease the probability of exit and increase the probability of ownership change. The effect of physical productivity on the probability of exit or ownership change is generally insignificant. Our results also show that firms with higher demand-specific factors have higher probability to expand whereas the average firm-level physical productivity turns out to be an insignificant determinant of firm expansion.

2. Industry Description

The modern broiler industry is a vertically integrated system of production, processing, and distribution. Broiler companies control all stages of production ranging from breeding flocks and hatcheries to broiler grow-out and processing. The finishing stage of production (the final stage of the production process where one-day-old chicks are brought to the farm and grown to market weight) as well as the production of hatching eggs (broiler breeder operations) are both organized almost entirely through contracts between processors and independent growers. Broiler companies typically run their operations through smaller divisions (profit centers) found throughout the country, but mainly in the South and Southeast. In addition to organizing live production, each of these profit centers typically operates a hatchery, a feed mill, a transportation unit and a processing plant (Vukina and Leegomonchai, 2006). As the result of this industry organization, the market for live broilers does not exist. Live broilers represent the main input into broiler processing and are acquired by the processing plants via some non-transparent internal transfer pricing scheme.

In the last several decades, the increase in the production of broiler meat in the U.S. came from both the increase in the slaughter weights of chickens and from the increase in the number of chickens grown. Compared to 4.97 billion chickens slaughtered in 1987, the total number in 2007 was 8.9 billion, which represents an increase of 79 percent. During the same time period the production grew from 15.5 billion pounds (ready-to-cook) to 36.2 billion pounds which represents an increase of 133.5 percent. In post-2008 period, the production stagnated and, more

recently, even decreased.[7] Three main factors contributed to this growth: growth in population, growth in per capita consumption, and exports. Table 1 shows the contributions of those factors in 5-year intervals between 1987 and 2007. The most significant contribution came from the growth in per capita consumption which was as high as 19% in the 1987-1992 period. Chicken consumption surpassed beef in 1993 and became the number one meat item consumed in the United States. The growth in the volume of production was accompanied by the decrease in price. Broiler retail price dropped from $1.16/lb in 1987 to $0.89/lb in 2007, whereas the prices of its substitutes, beef and pork, remained relatively stable.[8]

Another contributing factor for the expansion of the broiler industry's production was exports. From 1987 to 2007 the U.S. broiler exports increased approximately seven times, from less than 1 billion pounds to nearly 7 billion pounds. The ratio of exports to U.S. total production also increased from 4.87% in 1987 to 18.52% in 2010. Strong reliance on exports was essential for industry profitability. Domestic consumers' strong preference for white (breast) meat and the lack of demand for dark meat (leg quarters) forced the U.S. broiler industry to look to overseas markets to get rid of large quantities of leg quarters (Rabobank, 2011).

During the same period, the broiler industry has become more and more concentrated. The Herfindahl index (HHI) of poultry processing industry increased from 735 to 1224 from 1992 to 2007. The industry's top-4-firms concentration ratio increased from 44.24% in 1992 to 58.52% in 2007, and top-8-firms concentration ratio increased from 60.30% to 72.14%.[9] For comparison purposes, the rest of the meat complex in the U.S. is even more concentrated with beef and pork industries' concentration ratios exceeding that of the broiler industry. In 2007, the 4-firm concentration ratio of beef packers was 83.5% and of pork packers 66%, compared to only 58.5% for broilers (Hendrickson and Heffernan, 2007).

[7] The meat production data (ready-to-cook) comes from Read Meat and Poultry Production Data http://www.ers.usda.gov/data-products/livestock-meat-domestic-data.aspx#26056 and the number of heads slaughtered comes from Livestock and Poultry Slaughter data http://www.ers.usda.gov/data-products/livestock-meat-domestic-data.aspx#26063; last accessed January 16, 2014.

[8] Meat prices were calculated by first averaging the monthly prices to obtain yearly prices and then deflating the resulting yearly prices by meat consumer price index. Retail prices for broilers are prices for the whole birds. All data are available at USDA-ERS's website http://www.ers.usda.gov/data-products/meat-price-spreads.aspx#.Uai3tJyKIz4, last accessed on 5/31/2013.

[9] HHI was calculated using Census of Manufactures data where top 50 firms were used while the rest was put in the "other" category. Top-4-firms and top-8-firms are the top 4 firms' market shares and top 8 firms' market shares based on the value of shipments from the Census of Manufactures.

As mentioned in the introduction, the studied period was characterized by the intense mergers and acquisitions activity. For example, in 1998 Tyson and Hudson Foods merged, in 2000 Seaboard Farms was acquired by ConAgra Poultry and in 2002 BC Rogers was acquired by Koch Foods.[10] Most acquiring firms were large while most acquired firms were small. The exception to this is Gold Kist who was No. 3 when it was acquired by Pilgrim's Pride who was then No. 2.[11] The Pilgrim's Pride's case is compelling: after a series of aggressive acquisitions (Wampler Foods Inc. in 2001, ConAgra Poultry Co. in 2003, and Gold Kist in 2006), Pilgrim's filed for Chapter 11 bankruptcy protection on December 1, 2008, because of the deterioration of poultry pricing combined with an increase in input costs and the company's lack of liquidity to withstand the downturn[12].

Despite a substantial mergers and acquisitions activity, the rankings of the leading players remained relatively stable during the analyzed period (see Table 2). Tyson was and remained the number one U.S. broiler company, except briefly in the 2006-2007 period when its leading position was taken over by Pilgrim's. The other two large players are Perdue Farms and Sanderson Farms. Perdue represents an interesting case because it is the only privately-owned large company in this market segment which is dominated by publicly held companies. Table 2 also shows that the production shares of most broiler companies were stable over the time period except Pilgrim's. Pilgrim's production share increased from 7% in 1997 to 25% in 2009, peaking at 27% in 2006.

In the late 1990s and the first decade of 2000, the broiler industry grew three types of birds: 4 pounds, 5.5 pounds and 7 pounds live weight and used 4 marketing models; for details see Vukina (2005).[13] Plants generally specialize in processing of one size of birds only. Switching to producing and processing different size of birds requires re-alignment of production contracts

[10] References to specific companies in the text are based on public information. While most of the statistical/econometric analyses discussed later in the text employed confidential Census of Manufactures data references to individual companies by name are based on information that comes solely from an external trade publication, Watt Poultry USA (various issues).

[11] Pilgrim's sent an unsolicited proposal to Gold Kist offering to purchase all of the outstanding shares for $20.00 per share in cash on 18 August, 2006. The agreement was reached on 4 December 2006 at the price of $1 higher than Pilgrims' initial offer.

[12] Pilgrim's Pride successfully emerged from bankruptcy protection on December 28, 2009 after 64% of its stake was acquired by the Brazilian conglomerate JBS earlier that year.

[13] The study is based on the industry self-reported data obtained from AgriStats. AgriStats is a private company that partners with their customers to identify efficiency opportunities on a farm, flock, or plant level. They service chicken, turkey, commercial egg, and swine industries and utilize customized reports to identify for each customer exactly how every level of their operation performed in a given period, and how they compared to similar companies in the industry.

and re-calibration of processing lines and happens very infrequently. The small birds were processed and marketed either as WOG (without giblets which means whole birds) or as 8 piece cuts. As seen from Table 3, the industry average production cost in 2004, which consist of live production cost, processing and packaging cost and wholesale marketing cost, for WOG was 56.34 cents per eviscerated pound and 64.5 cents for 8-piece cuts. The medium size birds are always processed and marketed as tray pack and their average production cost was 66.9 cents. The large birds are mainly used in deboning and the industry average production cost was 60 cents per pound. A detailed analyzes of the industry structure reveals that in 2004, 6.5% of the industry output (by processed weight) was produced in WOG marketing channel, 20.9% in the small birds 8 piece cut, 35.2% in the medium size bird tray pack, and 37.4% in the large birds deboning channel. Using these shares one can aggregate WOG and 8-piece cut marketing channels to come up the weighted average production cost for a typical small bird plant in the amount of 62.6 cents per eviscerated pound. As seen from the bottom panel of Table 3, the average production costs across different plant sizes are not dramatically different. For three years (2002, 2003 and 2004) for which the data are available, the cost deviation of the small birds plants from the industry average is less than 1 percent, for large birds plants between 5 and 6 percent and for the medium birds plants between 6 and 9 percent, depending on the year.

Using the same data source for average prices received by plants for different product categories (whole birds, fast food 8-9 piece, deboned breast, parts, etc.) one can calculate the average revenue streams for different plants and see that they are also quite similar. As seen from the bottom panel of Table 3, in 2004 the average revenue per eviscerated pound for small plants was 73.1 cents, medium size plants 76.8 cents and the large plants 74.1 cents. Expressed as a deviation from the industry average revenue, small birds plants' revenues were off by between 2 and 5 percent, large birds plants' between 1 and 5 percent and medium birds plants' between 3 and 13 percent, depending on the year.

The above results reveal that the cost structure and the profitability of poultry firms (plants) who specialize in the production and processing of different size birds on average do not differ too much. On the cost side, this is mainly the result of the inverse relationship between live production and processing cost in the sense that heavier birds are more expensive to grow (because feed conversion deteriorates with weight) but relatively less expensive to process. On the revenue side, this is mainly caused by the fact that chicken processing is a fixed proportions

technology where it is impossible to engineer, say, a third drumstick from some lesser valued parts, and also by the fact that equal cuts (parts) are homogeneous and hence sold by weight alone.

3. Theoretical Framework

The empirical framework in this study is developed based on the theoretical model of Foster, Haltiwanger and Syverson (2008) which shows how idiosyncratic technology and demand-specific factors can jointly determine producers' long-run survival prospects in an industry equilibrium. In that model, consumers maximize utility by choosing over different varieties of products within the industry. Firms in the industry maximize profits by choosing the output price, given firm-specific demand curve, input prices and technology. Outsiders choose whether to pay the entry cost to take a draw of a combination of firm-specific demand, input prices and technology. After receiving a draw, they decide whether to enter the industry or not.

In equilibrium, the optimal price charged by a producer is positively related to demand-specific factors and input price and negatively related to technology-specific factors. Equilibrium quantity of good is positively related to demand-specific factors and technology-specific factors, but negatively related to input price. These results are intuitive because the stronger the representative consumer's preference for the product, the higher the price the producer could charge and the higher the equilibrium quantity. Higher input price and lower productivity will increase the marginal cost of the variety, which will induce the producer to increase the price to recover the cost and hence decrease the equilibrium quantity (supply curve shifts up). Another major implication of the model is that firms with higher demand-specific factors, higher productivity and lower input prices are more likely to survive.

4. Data Description

The data for this study comes from the Census of Manufactures (CMF) conducted by the U.S. Census Bureau twice a decade in years ending in "2" and "7". In particular, we use data from years 1987, 1992, 1997, 2002 and 2007. The CMF collects information on plants' annual value of shipments and total volume of shipments when feasible (for chicken processing plants it is the total quantity of shipments in thousands of pounds). This allows us to compute physical productivity and plant-level price for each plant in our dataset.

The rules for inclusion in the sample are summarized as follows. Poultry processing plants were first identified as plants with NAICS codes starting with 311615 (since 2002) or SIC codes starting with 201500 (before and up to 1997). This step resulted in approximately 500 plants in each of the four census years. For large[14] poultry processing plants, CMF reports their outputs by type: young chickens (3116151); processed poultry and small game products containing 20% or more poultry or meat (excluding soups) (311615D); turkeys (3116157); hens and fowl (3116154); and other poultry and small game (311615A). For small poultry processing plants, however, CMF reports different outputs all under one category (311615W). Because this study focuses on the broiler processing plants, we only kept plants whose revenue from young chickens (3116151) accounts for at least 90 percent of their total revenue. With this criterion, we were left with approximately 170 plants from each census year. The excluded plants are of the following types: about 170 plants produce processed poultry and small game products containing 20% or more poultry meat; about 60 plants process turkey; and about 100 plants are small plants and their outputs are reported under one category.[15] Finally, we dropped the plants with missing values on some or all of the variables required for the empirical analysis. The final sample size consists of about 110 plants from each census year.

Next, all observations in our final sample are classified into three groups. A plant-year observation belongs to the exiting group in year t (t equals 1992, 1997, 2002 or 2007) if the plant was closed or stopped processing broilers between year $t-4$ and year t (including year t). A plant-year observation belongs to the surviving group in year t if it continued to operate under the same ownership between year $t-4$ and year t (including year t). Finally, a plant-year observation belongs to the acquired group in year t if it was acquired by another owner between year $t-4$ and year t (including year t) and continued processing broilers. This variable will be used as the dependent variable in all our econometric models below.

5. Empirical Approach

[14] Large plants are plants with more than 250 employees and small plants as those with less than 250 employees.
[15] The exclusion of small plants from our sample could raise some concerns, but the inclusion of these plants is neither feasible nor necessary. It is not feasible because their outputs are reported under one category and hence there is no way for us to know how much young chickens these plants produced. Also, it is not necessary to include them in the sample because the small plants, as a whole, only account for a small percentage of the total industry value of shipments (about 0.25% in 2002).

The empirical approach used in the paper critically hinges on the assumption that the chicken slaughter plants produce a homogenous product. If this were not the case, then revenue based total factor productivity (TFP) measure would be more appropriate than the quantity based TFP. This is because the production of, say, better quality product requires more inputs to produce the same quantity of output. Therefore, if one uses physical output as the dependent variable in the TFP estimation, as we do here, in case of pronounced product heterogeneity the TFP will be underestimated. Instead, if one uses the revenue based TFP measure, then one can no longer separate the TFP from the demand specific factors because revenue depends on price and price depends on demand specific factors. However, the plant-level homogeneity assumption does not imply the complete absence of firm-level brand loyalty even for otherwise homogenous products. In fact, to the extent that some degree of brand loyalty exists, it becomes one of the demand specific factors in the demand model.

The assumption that the industry, as defined in this paper, produces technologically homogenous output is, in light of the existing literature, controversial but nevertheless quite plausible for two reasons.[16] First, of five output categories reported by the Census, we only use plants whose revenue from "young chickens" accounts for at least 90% of their total revenue. This means that we generally excluded plants that are engaged in higher level of processing, in particular category "processed poultry and small game products containing 20% or more poultry meat", where product differentiation surely becomes more pronounced. Second, the similarity of plant level costs and revenues per pound of processed weight, based on the highly reliable self-reported industry data, indicates that chickens slaughter plants are reasonably homogenous.

5.1. Measuring Productivity

[16] Ollinger, MacDonald and Madison (2005) used the same Census data for partially overlapping time period (1972-1992) and found that the best estimated cost function includes controls for product mix and that higher processed products impose higher cost. Ollinger (2011) used the Census data thorough 2002 and also found that product mixes varied across plants and that more processed products raised plant-level cost. There are at least two reasons for the discrepancy between these claims and ours. First, the definition of the sample in the above studies in terms of which plants are included is different than ours. Their sample includes plants that process other poultry products such as hens and ducks, as well as plants that produce further processed chicken products like chicken ham and sausages. Our sample only includes those plants that produce minimally processed broiler meat. Secondly, controlling for product mix relying on the output category classification systemused by Census (wet ice pack, dry ice pack, tray pack, etc.) is problematic because these categories are associated with packaging and not with the level of processing. In fact, Ollinger at al (2005) are fully aware of this problem and argued that Census output categories for beef and pork were more meaningful than those for chickens.

We measure plant-level productivity using the concept of total factor productivity (TFP). With Cobb-Douglas production technology, TFP can be written as follows,

(1) $$tfp_{it} = y_{it} - \alpha_{lt}x_{l,it} - \alpha_{kt}x_{k,it} - \alpha_{mt}x_{m,it} - \alpha_{et}x_{e,it},$$

where y_{it} and x_{it}s are the natural logarithms of physical output, labor, capital, materials and energy inputs and, under the assumption of constant returns to scale, αs are output elasticities of the corresponding inputs.[17] Physical output (y_{it}) consists of *processed chickens*, which include the following products defined with 10-digit census product codes: wet ice pack broilers and fryers (3116151111); dry ice pack broilers and fryers (3116151221); tray pack broilers and fryers (3116151331); other broilers and fryers (3116151441); roasters and capons (3116151551). This variable is measured in thousands of pounds.

There are four important inputs in broiler production: materials, labor, energy and capital. Input quantities (x_{it}s) and prices (w_{it}s) were constructed as follows. Materials quantity, $x_{m,it}$, is measured as the total quantity of live broilers (in thousands of pounds) used in production (census material code 112320015: young chickens). The price of the materials (in nominal dollars per pound), $w_{m,it}$, is computed by dividing the expenditure on materials (in thousands of dollars), which include expenditure on live broilers, boxes and other packaging materials, by $x_{m,it}$. The labor price (in nominal dollars per hour), $w_{l,it}$, is calculated by dividing the expenditure on production workers (in thousands of dollars) (which includes annual payroll and fringe benefits for both leased and non-leased workers) by the total production workers' production hours. The labor quantity (in thousands of hours), $x_{l,it}$, is computed by dividing the sum of the expenditure on production workers and non-production workers (in thousands of dollars) by $w_{l,it}$. The implicit assumption here is that production workers and non-production

[17] We rely on the constant returns to scale assumption because it is plausible and analytically convenient since it enables the exact computation of the TFP instead of a cumbersome estimation plagued by severe endogeneity problems (see below). However, Ollinger, MacDonald and Madison (2005) found large and unexploited increasing returns to scale, even among relatively large poultry plants. Their result is puzzling for reasons well explained by the authors themselves (p. 124). Expanding processing volumes require increased production of live broilers which can be accomplished either by increasing the geographic coverage of the complex, which increases the transportation cost and mortality (due to transport related heat exhaustion), or increasing the density of production, which is accompanied by increased environmental problems (manure disposal) and biosecurity risks. Therefore, the only way the increasing returns to scale at the processing level can be reconciled with the constant returns to scale at the complex level is by implicitly asserting that poultry companies are managed by incompetent executives who regularly build oversized processing plants knowing full well that they will never produce enough live birds to run them on the efficient (minimum average cost) scale.

workers have the same average wage rate.[18] The energy price (in nominal dollars per kilowatt), $w_{e,it}$, is calculated by dividing the expenditure on electricity (in thousands of dollars) by the quantity of electricity used (in kilowatts). The energy quantity (in thousands of kilowatts), $x_{e,it}$, is computed by dividing the sum of the expenditure on electricity and fuel (in thousands of dollars) by $w_{e,it}$. The implicit assumption here is that the electricity price and the fuel price are the same when electricity and fuel are converted into the same energy units (the amount of heat generated by one kilowatt of electricity).[19] Finally, prices for material, labor and energy are then deflated by CPI (2005=100).

The calculation of the rental price of capital and the quantity of capital is more complicated.[20] The 3-digit industry level (in this case NAICS=311: Food, Beverage and Tobacco Products) rental price of capital, $w_{k,t}$, can be computed by dividing industry level capital income (KY_t, in nominal billions of dollars) by the 3-digit industry level capital input (in nominal billions of dollars). KY_t is directly available from the multifactor productivity database published by the Bureau of Labor Statistics (BLS).[21] Capital input is not directly available, but can be constructed using data on productive capital (PK_t, in deflated billions of dollars), capital composition (KC_t, a ratio, unit free) and price index of investment ($PIINV$). PK_t and KC_t are directly available from the same BLS table, and $PIINV$ is available from NBER.[22] As PK_t is the deflated aggregate industry level productive capital and KC_t is the ratio of industry level capital input in 2005 dollars to industry level productive capital in 2005 dollars, $PK_t * KC_t/100$ gives us the industry level capital input in 2005 dollars. Multiplying $PK_t * KC_t/100$ by the corresponding $PIINV$ then gives us the industry level capital input in nominal dollars. For example, $\frac{PK_t * KC_t * PIINV92}{100}$ gives us the industry level capital input in 1992 dollars. In summary, we calculate the industry level rental price[23] for a particular year (e.g. 1992) as,

(2) $$w_{k,t} = \frac{KY_t}{PK_t * \frac{KC_t}{100} * PIINV92}.$$

[18] This assumption is necessary as the Census data does not report the total hours for non-production workers. An alternative assumption could be to assume the wages for non-production workers are 150% of those for production workers (Kehrig 2012).
[19] This assumption is necessary as the Census data does not report the quantity of fuel used.
[20] We adapted the method used in Kehrig (2012).
[21] Bureau of Labor Statistics: "Capital by Asset Type for NIPA-Level Manufacturing Industry," available at: http://www.bls.gov/mfp/mprdload.htm, last accessed on 5/31/2013.
[22] The deflator is from the NBER-CES Manufacturing Industry Database, available at http://www.nber.org/nberces/, last accessed on 5/31/2013.
[23] Although we call it a "price," it is just a ratio of rental expenditure to market value of the capital.

The quantity of capital (in thousands of nominal dollars), $x_{k,it}$ consists of two parts: assets owned by the plant, and assets rented from others. Assets owned by the plant is reported in the Census data as total asset at the beginning of the year (TAB_{it}, in thousands of nominal dollars). Because TAB_{it} is the book value (the historical cost of the capital minus the aggregate depreciation), we multiply it by $\frac{NKC_t}{GKH_t}$ to transform it into market value (i.e., how much the firm would get if it sells the capital in the market). NKC_t is the industry level net capital stock in nominal dollars (1992 for example) and GKH_t is the industry level gross capital stock in book values, both of which are available from the capital stock data published by the Bureau of Economic Analysis (BEA).[24] Assets rented from others is computed by dividing total rental payment (TR_{it}), which is directly available from CMF (in thousands of nominal dollars), by the industry level rental price $w_{k,t}$. These two parts are then summed together to obtain the quantity of capital stock, that is,

$$(3) \qquad x_{k,it,nominal} = TAB_{it} * \left(\frac{NKC_t}{GKH_t}\right) + \frac{TR_{it}}{w_{k,t}}.$$

The capital quantity in (3) is in nominal dollars and can be used with the nominal rental price $w_{k,t}$ to obtain the nominal capital expenditure. Finally, we deflate the nominal capital quantity in (3) using $PIINV$ from the corresponding year to get the real capital input quantity, which is the one we used in the productivity calculation. For example, for $t = 1992$, $x_{k,it}$ is computed as,

$$(4) \qquad x_{k,it} = \frac{TAB_{it} * \left(\frac{NKC_t}{GKH_t}\right) + \frac{TR_{it}}{w_{k,t}}}{PIINV92}.$$

We now turn to discussion of elasticities, αs, in (1). The traditional way of obtaining the elasticities would be to estimate a production function by regressing the output y_{it} on the inputs x_{it}s. However, as pointed out by Olley and Pakes (1996), this approach may yield biased estimates because certain inputs like labor, which can be adjusted in the short run, may reflect plants' responses to unobserved technology shocks and hence these input variables are likely to be endogenous in the production function regression. Two solutions are proposed in the literature to overcome this problem. The first is to use the instrumental variable approach and the second is to use the control function approach (e.g. Olley and Pakes 1996; Levinsohn and Petrin 2003).

[24] See Tables 3.1ES and 3.3ES on Bureau of Economic Analysis's webpage: http://www.bea.gov/iTable/iTable.cfm?ReqID=10&step=1#reqid=10&step=1&isuri=1, last accessed on 5/31/2013.

The idea of the latter approach is to use another variable as a proxy to control for the unobserved plant technology shocks that are likely to influence plants' short-run input decisions.

Both approaches are not feasible in our context. First, CMF only collects data on a limited number of variables and no variables are good candidates to use as instruments for potentially endogenous variables. Second, the key assumption for the control function approach to be valid is that there is a monotonic relationship between the proxy variable, which is usually an input variable that can be adjusted in the intermediate run like materials, and the unobserved plant technology shock. But the main message of this study is that in addition to plant level productivity, plant level idiosyncratic demand factors may also play an important role in plant decisions. In other words, plants' decisions including the decisions on the proxy variables are functions of both unobserved plant level technology shock and unobserved plant level demand shock. In this case, the assumption that there is a monotonic relationship between the proxy variable and the unobserved technology is unlikely to be true.

Instead of estimating a production function, we follow Foster, Haltiwanger and Syverson (2008) and compute rather than estimate αs. Given the Cobb-Douglas production function and assuming constant returns to scale, αs can also be interpreted as input expenditure shares. For each plant, we calculate expenditures for materials, energy and labor by multiplying nominal prices (w_{it}s) with their corresponding quantities (x_{it}s). Capital expenditure is calculated as the product of the industry level rental price ($w_{k,t}$) and the plant's nominal capital quantity ($x_{k,it}$). With these, we can compute the input expenditure shares for each plant in each census year in our sample. The averages across all plants in a particular census year are then used as the estimates of αs for that year. The implemented approach has two other advantages over the estimation approach, even if the estimation approach were feasible. First, control function approaches, like in Olley and Pakes (1996) and Levinsohn and Petrin (2003), require the use of at least 1-period lagged variables in estimation, reducing the sample size and hence the efficiency of the estimates. Second, in econometric estimation approach, the α parameters in (1) are usually constrained to be the same across different years in order to reduce the number of parameters to be estimated, while the computation approach allows them to vary across years and hence is more general.

The left panel of Table 4 reports the summary statistics for the calculated TFP variable by the plant category. The actual numbers are not interpretable but their magnitudes are comparable

in the sense that higher numbers mean higher TFP. Though observations in the survival group have lower average TFP relative to exiting and acquired groups, all the differences are statistically insignificant.

5.2.Measuring Idiosyncratic Demand Factors

To construct the measure of plant-specific idiosyncratic demand factors, we estimate the following demand function,

$$\ln y_{fit} = \alpha_0 + \alpha_1 \ln p_{fit} + \sum_t \beta_t Year_t + \sum_f \gamma_f Firm_f + \mu_{fit}, \quad (5)$$

where y_{fit} is the physical output of firm f's plant i in year t, p_{fit} is the plant-level output price in year t,[25] $Year_t$s are yearly dummies and $Firm_f$s are firm dummies.[26] Yearly dummies are included in the regression to account for yearly (macroeconomic) shocks in demand for chicken meat. Firm dummies are included in the regression to account for firm-specific factors (e.g. firm reputation, networks, political connections, etc.) that can potentially influence the demand for a plant's output. Because plant-level price is correlated with the plant-specific idiosyncratic demand factors (see Section 3), we use the instrumental variables approach to correct for the endogeneity bias. In our context, TFP and input prices are good candidates for valid instruments. The theoretical model implies that TFP is negatively correlated with output price and input prices are positively correlated with output price, see Section 3. At the same time, these variables are supply shifters, which are not correlated with the demand shocks μ_{fit}.[27]

We estimate (5) using 2SLS with TFP and deflated materials price (deflated $w_{m,it}$), as the instruments. Wages and energy price were not used as instruments because we found that they are not highly correlated with the plant-level output price, the endogenous variable in question. The price of capital is not used because it is only observed at the industry level, which makes it perfectly collinear with the yearly dummies. The estimated term $\sum_f \hat{\gamma_f} Firm_f + \hat{\mu}_{fit}$ from the demand estimation represents the sought after measure of plant-specific idiosyncratic demand factors for firm f's plant i in year t.

[25] Plant-level output price was computed as the plant-level value of shipments divided by the plant-level output quantity, both of which are directly available from CMF.

[26] Since the number of plant-year observations in our data set is relatively small, to alleviate the incidental parameters problem, we only include firm dummies for firms who have more than 4 plant-year observations in our sample.

[27] Using supply shifters to identify the demand function is a standard econometrics textbook solution for solving the endogeneity problem in a supply and demand simultaneous equation system.

Results from the 2SLS regression are reported in Table 5. The first-stage F test rejects the hypothesis that the instruments used are weak and the Basmann's (1960) overidentification test cannot reject the hypothesis that the instruments are exogenous. The price coefficient is negative and statistically significant. The magnitude of the coefficient of -1.32 is consistent with the idea that consumers exhibit low or no brand loyalty when it comes to buying chicken meat from different plants/firms.[28] Using the estimated coefficients, idiosyncratic demand factors for each plant-year observation in the sample are calculated. The summary statistics are reported in the right panel of Table 4. It can be seen that observations in the acquired group have the highest average idiosyncratic demand factors, while observations in the exit group have the lowest. This implies that plants with lower idiosyncratic demand factors are less likely to survive or be acquired by other firms, as expected. On the other hand, plants with higher idiosyncratic demand factors are more likely to become targets of mergers and/or acquisitions.

6. Determinants of Selection

With the calculated TFP and the estimated idiosyncratic demand factors for each plant-year observation, we can now examine the effects of these two variables on plant and firm performance. The two-step approach involves using plant-level productivities and demand specific factors from the first step as explanatory variables in the discrete choice models in the second step. It is well known that in this kind of two-step analysis, the standard errors from the second-step regressions are biased. The correct standard errors are obtained using the bootstrap procedure with 1,000 iterations. Below, we first examine the effects of productivity and demand specific factors on plant survival and ownership change and then examine their effects on firm expansion.

6.1. Plant-Level Analysis

We first examine the effects of TFP and idiosyncratic demand factors on plant exit using the simple Logit regression.[29] The dependent variable equals to 1 if the plant-year observation belongs to the exit group and 0 otherwise. The independent variables used in the regression include the two variables of interest, capital stock ($x_{k,it}$ defined above) and yearly dummies. The

[28] USDA-ERS maintains a database of demand elasticities from the literature, available at http://www.ers.usda.gov/Data/Elasticities/, accessed on 5/31/2013. The five aggregate level own-price chicken elasticities range from −0.3 to -1.13, all smaller (in absolute value) than our plant level elasticity result. This result is reasonable since the demand at the aggregate level is typically less elastic than the demand at the disaggregate level.
[29] Results from the Probit regression are very similar.

purpose of controlling for the capital stock variable is to distinguish the effects of short-run from long-run productivity. The TFP variable calculated above reflects a plant's short-run productivity, while a plant's capital stock reflects its long-run productivity (Olley and Pakes 1996). The capital variable could also be interpreted as controlling for the plant size.[30] Results are collected in Table 6. The parameter estimates show that higher productivity (both short-run and long-run) increases the probability of exit, but the effects are statistically insignificant. In contrast, higher idiosyncratic demand factors have a significant and negative impact on the probability of exit. The marginal effects of a one unit increase in each independent variable are also reported in the right panel of Table 6. Results show that a one unit increase in the idiosyncratic demand factors corresponds to a decline in exit probabilities of 8.27 percentage points, an economically significant effect.

Next, we examine the effects of TFP and idiosyncratic demand factors on plant ownership change. The dependent variable in this regression equals to 1 if the plant-year observation belongs to the acquired group and 0 otherwise. Results are collected in Table 7. Again, only the idiosyncratic demand factors variable has a significant impact and a one unit increase in the idiosyncratic demand factors increases the probability that the plant will be acquired by another firm by 10.38 percent.

Finally, as a robustness check, we re-estimate the effects of TFP and demand specific factors on the probabilities of plant exit and ownership change jointly using the multinomial Logit model. In this regression, the dependent variable equals 1, 2 and 3 if the plant-year observation belongs to the surviving, exit and acquired groups, respectively. The surviving group is the baseline group. Results from this regression are reported in Tables 8 and 9. Our main results continue to hold, that is, only the idiosyncratic demand factors variable has a significant impact and it decreases the chance of plant exit and increases the chance of plant ownership change. More specifically, Table 9 shows that a one unit increase in the idiosyncratic demand factors variable decreases the probability of exit by 8.81% (significant at 5% level) and increases the probability for the plant to be acquired by another firm by 10.23% (significant at 1% level), while has almost no impact on the probability of continuing operation by the same owner (survival).

[30] As a robustness check, we also estimated the same econometric models without the capital variable. Results from these regressions are reported as model (2). In all regressions, the two sets of results are very similar and hence we focus our discussion on the regressions with the capital variable.

At first glance, the above results differ markedly from some of the earlier studies, e.g. McGuckin and Nguyen (1995) and Nguyen and Olinger (2006), which showed that labor productivity is positively associated with the ownership change and the subsequent productivity growth. However, these studies define labor productivity as the value of output divided by the number of employees, which is a revenue based productivity measure. As explained earlier, this type of revenue based productivity measure confounds the effects of physical productivity and demand-specific factors. If the true underlying driving force of industry dynamics is demand specific factors rather than physical productivity as we found, then our results are in fact consistent with those earlier studies because higher demand specific factors lead to higher output prices and hence higher revenue based productivity. Disentangling the two effects was exactly the central motivation of this study.

6.2. Firm-Level Analysis

As mentioned before, the significant increase in the total production of poultry industry during the analyzed period came about via the increased number of birds grown but even more so via the increase in average live weight. At the same time, the number of firms in the industry remained relatively stable and the increase in the number of plants was only modest. As seen from Table 2, the number of plants owned by the top-10 ranked poultry companies increased from 106 in 1997 to 119 in 2007, with the peak of 122 in 2006. During the same time period, among 165 firms in our data set, 27 of them expanded their production by either opening a new plant or by acquiring an existing plant (see Table 10). In order to explain this activity, in this section we study the effects of TFP and idiosyncratic demand factors on firm expansion. For the purposes of this analysis, we aggregate plant-level data to obtain firm-level data. The firm-level output is simply the sum of plant-level outputs, while other firm-level variables are the weighted averages of plant-level variables. For TFP and output price, we use plant output as the weight. For each input price, we use the corresponding plant input as the weight.

A firm-year observation belongs to the expanding group in year t if it opened or acquired at least one new plant between year $t-4$ and year t (including year t). The summary statistics for the firm-level TFP are reported in the left panel of Table 10, which shows that expanding firms have slightly lower productivity than non-expanding firms. To construct the firm-level idiosyncratic demand factors, we estimate the following firm-level demand function,

(6) $$ln\, y_{ft} = \alpha_0 + \alpha_1 ln\, p_{ft} + \sum_t \beta_t Year_t + \gamma Single_{ft} + \mu_{ft},$$

where y_{ft} is the firm-level output quantity, p_{ft} is firm-level output price, $Single_{ft}$ is a dummy variable identifying single-plant firms, and μ_{ft} is an error term. Similar to the plant-level analysis, we use firm-level TFP and materials input price as the instrumental variables for the endogenous price variable. Estimation results from the 2SLS are reported in Table 11, which are similar to plant-level demand estimation results. The instruments pass both the overidentification test and the first-stage F test for weak instruments. The demand for chicken is elastic and slightly less elastic than the demand for chicken at the plant level. This is reasonable as demand at disaggregate level is usually more elastic than demand at aggregate level. Also, the results show that demand is lower for single-plant firms. This is consistent with the idea that single-plant firms are usually small local firms and spend less on demand-enhancing marketing activities. The estimated term $\hat{\gamma} Single_{ft} + \hat{\mu}_{ft}$ from the demand estimation is our measure of firm-level idiosyncratic demand factors for firm f in year t. The summary statistics of this variable are reported in the right panel of Table 10. As we can see, expanding firms clearly have higher idiosyncratic demand factors.

With the calculated TFP and the estimated idiosyncratic demand factors for each firm-year observation, we can now examine the effects of these two variables on firm expansion. Again, we use the simple Logit regression and correct standard errors are obtained using the bootstrapping method. The dependent variable equals 1 if the firm-year observation belongs to the expanding group in year t and 0 otherwise. Estimation results are reported in Table 12. In terms of parameter estimates, both idiosyncratic demand factors and TFP have a positive but insignificant effect on the chance for the firm to expand. In terms of marginal effects, however, higher idiosyncratic demand factors increases the chance for the firm to expand and this effect is statistically significant at 1% level. The effect of TFP remains to be statistically insignificant. In terms of magnitude, a one unit increase in idiosyncratic demand factors leads to an increase in the probability of expansion by 8.07%.

7. Conclusions

Most of the empirical literature using business-level data points to within-industry reallocation, and its associated entry and exit of firms, as the most important determinant of industry performance. The effect of this dynamics on aggregate productivity has received

particular attention. The important mechanism driving aggregate productivity movements is the reallocation of market shares to more efficient producers, either through market share shifts among incumbents or through entry and exit. Low productivity plants are less likely to survive and thrive than their more efficient counterparts. Hence, the productivity-survival link becomes a crucial driver of productivity growth.

In reality, the productivity-survival link is an over-simplification because what drives the within-industry selection is not the productivity *per se* but the profitability, where productivity is only one of the several factors that determines profits. Most of the earlier literature in this field relied on the so called traditional productivity measure (revenue divided by a common industry-level deflator as a measure of output). The problem is that traditional productivity (revenue-based productivity) confounds the effects of physical productivity and demand-specific factors on market selection. The traditional productivity measure uses revenue as a measure of output, while physical productivity uses real output (i.e., the traditional productivity measure is a product of price and physical productivity). Price is positively correlated with demand-specific factors, because the stronger the consumers' preference for the product, the higher the price that a producer can charge. Therefore, higher traditional productivity does not necessarily mean higher physical productivity. Higher traditional productivity could be a result of the entire host of idiosyncratic factors influencing firm's demand, all of which can increase the probability of the firm's survival. Therefore, using traditional productivity measure to study the effect of productivity on market selection is flawed because it ignores the effects demand-specific factors.

In this article we study the effects of physical productivity and demand-specific factors on plants' and firms' performance in the U.S. broiler industry. There are several reasons why poultry processing industry presents itself as a good candidate to study the importance of demand specific factors on industry dynamics. First, the industry can be considered as reasonably homogenous as indicated by the similarity of average operating cost and revenues across typical plants. Second, the industry is vertically integrated where firms control the entire chain from growing live birds to processing and marketing of chicken meat. The grow-out of live chickens is contracted with independent farmers where the control of all essential inputs (feed and genetics) is retained by the companies. All these features, combined with the fact that the processing stage of production is highly standardized and automated, point to possibly very small differences in physical productivity across firms. On the other hand, producer-level output prices vary

considerably among firms in the industry. Only small portion of products in manufacturing industries can be identified as homogenous products and the detailed production data availability limits researchers to generalize these results.

The use of the Census data enabled the observation of plant-level quantities; hence, we could separately measure physical productivity and demand-specific factors. As anticipated, the influence of physical productivity turns out to be an insignificant determinant of industry dynamics either at the plant-level or at the firm-level. This means that previous studies of other industries using traditional productivity measures could have overstated the importance of productivity on selection of plants, especially in narrowly-defined industries. On the other hand, plants with higher demand-specific factors are less likely to exit, more likely to survive, and more likely to be acquired. The same is true for firms where those with higher demand-specific factors are more likely to expand. The extension of the analysis of industry dynamics by measuring the demand-specific factors also shows that treating acquired plants the same as exiting or surviving plants is not appropriate. The results strongly indicate that acquired plants have significantly higher demand-specific factors than either the surviving or exiting plants.

The obtained results have compelling economic policy implications. The overall economic position of the U.S. broiler industry changed dramatically in the post 2008 recession period when the growth came to a halt. In this recent period, the industry has been facing significant challenges, the most serious one came from the high cost of animal feed, notably corn and soybeans. In addition, the domestic meat market was beginning to mature and the international competition was intensifying. The U.S. per capita consumption of broilers has been decreasing since 2006 and similar trends characterize other meats. The weakening of the demand for U.S. poultry in once traditional export markets such as Russia (mainly due to increased Russia domestic production) and the stiffer competition from the low cost producers (such as Brazil) required the broiler companies to reduce production. More consolidation, via industry exits, mergers and acquisitions, is a distinct possibility.

The cessation of industry expansion presents new challenges not only the companies but also the contract growers. Most growers already operate in non-competitive markets for their services, with very few integrators in any given area. Further consolidation caused by slow growth can deter the entry of new growers into the industry altogether and can potentially bias the competition outcomes in favor of big players. Since our results clearly show that demand

specific factors explain firms' survival and expansion, together with the fact that these demand specific factors, such as name recognition, superior market access, networking, political and business ties, etc., all favor large companies, the existing and potential new growers should quickly realize that their future is much more secure with large, internationally recognized, companies. Consequently, they would be reluctant to sign new contracts with smaller producers and eager to switch over to larger producers if those contracts are available in their region. This could expedite the industry concentration processes already well in place and could raise new concerns over the exercise of market power and its effect on contract growers' income and ultimately consumers' welfare.

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Table 1: Growth in Broiler Production (%), 1987-2007

Year Range	1987-1992	1992-1997	1997-2002	2002-2007
Broiler production growth rate	35.62	29.36	17.94	12.16
Contributions from growth in:				
Population	5.80	6.24	5.60	4.77
Per Capita Consumption	18.88	9.01	12.89	5.67
Other (mainly exports)	10.94	14.11	-0.55	1.72

Source: USDA-ERS all meat supply and disappearance data, available at: http://www.ers.usda.gov/data-products/livestock-meat-domestic-data.aspx#.UajABZyKIz4, last accessed on 5/31/2013.

Table 2: Leading US Broiler Companies 1997-2007

	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share	Rank	Plant	Share
	2007			2006			2005			2004			2003			2002			2001			2000			1999			1998			1997		
Pilgrim's Pride	**1**	38	25%	**1**	38	27%	**2**	27	17%	**2**	26	17%	**2**	26	16%	**3**	12	9%	**3**	13	9%	**5**	10	7%	**4**	9	7%	**4**	9	7%	**4**	9	7%
Tyson Foods	**2**	35	21%	**2**	39	22%	**1**	39	22%	**1**	40	23%	**1**	41	23%	**1**	41	24%	**1**	42	24%	**1**	42	26%	**1**	42	27%	**1**	42	26%	**1**	36	23%
Perdue Farms	**3**	10	8%	**3**	10	8%	**4**	10	6%	**4**	10	8%	**4**	11	8%	**5**	12	8%	**5**	13	8%	**4**	13	8%	**3**	13	8%	**3**	13	9%	**3**	13	9%
Sanderson Farms	**4**	8	6%	**5**	7	5%	**6**	7	4%	**5**	6	5%	**6**	6	4%	**7**	6	4%	**7**	6	4%	**7**	6	4%	**7**	6	4%	**8**	6	3%			
Wayne Farms	**5**	8	5%	**4**	8	5%	**5**	8	4%	**6**	8	4%	**5**	8	4%	**6**	8	5%	**6**	8	5%	**6**	8	5%	**6**	8	4%	**6**	8	5%	**7**	8	5%
Mountaire Farms	**6**	3	4%	**6**	3	4%	**7**	3	4%	**7**	3	4%	**7**	3	3%	**8**	3	3%	**10**	3	3%	**9**	3	3%									
House of Raeford	**7**	6	4%	**7**	5	3%	**10**	4	2%				**10**	4	2%																		
Keystone Foods	**8**	3	3%	**8**	3	3%																											
Koch Foods	**9**	6	3%	**9**	4	3%																											
O.K. Foods	**10**	2	2%							**9**	2	2%																					
Gold Kist							**3**	11	9%	**3**	11	10%	**3**	11	10%	**2**	12	10%	**2**	12	10%	**2**	12	10%	**2**	12	10%	**2**	12	11%	**2**	12	10%
ConAgra Poultry																**4**	13	8%	**4**	13	8%	**3**	13	9%	**5**	9	6%	**5**	9	6%	**5**	8	6%
Cagle's																															**6**	7	6%
Foster Farms													**9**	4	2%	**9**	4	3%	**8**	5	4%	**8**	6	3%	**8**	5	3%	**7**	5	3%	**8**	4	3%
Hudson Foods				**10**	5	2%	**8**	6	3%	**8**	6	3%	**8**	6	2%	**10**	6	3%	**9**	6	3%	**10**	5	3%	**9**	4	3%	**9**	4	3%	**9**	5	3%
Seaboard Farms																									**10**	4	3%				**10**	4	3%
Townsends																												**10**	4	3%			
Peco Foods							**9**	4	3%	**10**	4	2%																					
Total US production in millions of lb(1)			36,159			35,500			35,365			33,746			33,746			32,240			31,266			30,495			29,741			27,863			27,271

Source: Watt Poultry USA, annual surveys

(1) Total US figures are from Read Meat and Poultry Production Data http://www.ers.usda.gov/data-products/livestock-meat-domestic-data.aspx#26056

Table 3: Avearge Costs and Revenues for Representative Broiler Plants

	2002	2003	2004
Small-WOG (% share in total industry output)	8.9%	8.9%	6.5%
Live weight (Lb)	4	4	4
Yield	68.00%	68.50%	69.23%
(1) Live cost	0.2423	0.2563	0.2815
(2) Processing & Packaging Cost	0.1293	0.1303	0.1181
(3) Wholesale Marketing Cost	0.0371	0.0355	0.0387
Total Cost (1+2+3)	0.5227	0.5400	0.5634
Revenue	0.5379	0.5839	0.6882
Small - 8 piece cut (% share in total industry output)	24.5%	24.5%	20.9%
Live weight (Lb)	4	4	4
Yield	68.00%	68.50%	69.23%
(1) Live cost	0.2423	0.2563	0.2815
(2) Processing & Packaging Cost	0.1945	0.1991	0.1997
(3) Wholesale Marketing Cost	0.0371	0.0355	0.0387
Total Cost (1+2+3)	0.5879	0.6088	0.6450
Revenue	0.5672	0.6407	0.7465
Medium - Tray pack (% share in total industry output)	27.7%	27.7%	35.2%
Live weight (Lb)	5.5	5.5	5.5
Yield	71.1%	71.5%	72.64%
(1) Live cost	0.2408	0.2555	0.2830
(2) Processing & Packaging Cost	0.2218	0.2331	0.2380
(3) Wholesale Marketing Cost	0.052	0.0529	0.0414
Total Cost (1+2+3)	0.6124	0.6433	0.6690
Revenue	0.6369	0.7422	0.7684
Large - Deboning (% share in total industry output)	38.9%	38.9%	37.4%
Live weight (Lb)	7	7	7
Yield	72.80%	73.40%	74.08%
(1) Live cost	0.2453	0.2643	0.2906
(2) Processing & Packaging Cost	0.1783	0.1693	0.1741
(3) Wholesale Marketing Cost	0.0254	0.0248	0.0337
Total Cost (1+2+3)	0.5407	0.5542	0.6001
Revenue	0.5552	0.627	0.7412

		2002	2003	2004	2002	2003	2004
Small Bird Plants		plant averages			% deviations		
	Cost	0.5705	0.5904	0.6257	0.0%	0.1%	0.9%
	Revenue	0.5594	0.6256	0.7310	3.4%	5.0%	2.3%
Medium Bird Plants							
	Cost	0.6124	0.6433	0.6690	-7.3%	-8.9%	-6.0%
	Revenue	0.6369	0.7422	0.7684	-10.0%	-12.7%	-2.7%
Large Bird Plants							
	Cost	0.5407	0.5542	0.6001	5.2%	6.2%	5.0%
	Revenue	0.5552	0.627	0.7412	4.1%	4.8%	1.0%
Industry Total							
	Cost	0.5705	0.5910	0.6313	-	-	-
	Revenue	0.5792	0.6584	0.7484	-	-	-

Notes: Live costs are expressed in $/Lb of live weight; all other costs and revenues are expressed in $/Lb of eviscerated weight.
Percent shares of different type of plants are calculated based on the industry output expressed in pounds of eviscerated weight.

Source: Vukina (2005).

Table 4: Summary Statistics for Plant-Level TFP and Idiosyncratic Demand Factors

Type	N	TFP		Idiosyncratic Demand Factors	
		Mean	Std. Dev.	Mean	Std. Dev.
Exit	59	2.7297	1.26	0.0557	1.07
Survive	359	2.5468	1.02	0.5950	0.71
Acquired	53	2.7243	0.92	0.9076	0.46
Total	471	2.5904	1.04	0.5626	0.77

Table 5: Plant-level Demand Estimation Results using 2SLS

Variables	Estimate	Standard Error	P-value
Constant	10.5453***	0.1010	<0.0001
Log price	-1.3179***	0.2237	0.0003
Year dummies	included		
Firm dummies	included		
Tests		F-value	P-value
Basmann's overidentification test		0.62	0.4297
First Stage F-test for Weak IVs		138.36	<0.0001

***: significant at 1% level.

Table 6: Effects of TFP and Idiosyncratic Demand Factors on Plant Exit: Logit Model

Variables	Parameter Estimates		Marginal Effects	
	(1)	(2)	(1)	(2)
Constant	-2.2004*** (0.60)	-2.1211*** (0.51)		
TFP	0.1936 (0.15)	0.1884 (0.15)	0.0196 (1.51E-2)	0.0191 (1.51E-2)
Idiosyncratic Demand Factors	-0.8269*** (0.21)	-0.7826*** (0.17)	-0.0839*** (2.00E-2)	-0.0795*** (1.68E-2)
Capital	2.9508E-6 (7.80E-6)		2.9929E-7 (7.62E-7)	
Year dummies	included	included		

Notes: Standard errors are in parentheses. Marginal effects are effects of a 1 unit increase in the corresponding independent variable. * means significant at 10% level and *** means significant at 1% level.

Table 7: Effects of TFP and Idiosyncratic Demand Factors on Plant Ownership Change: Logit Model

Variables	Parameter Estimates		Marginal Effects	
	(1)	(2)	(1)	(2)
Constant	-2.7696*** (0.67)	-2.8439*** (0.60)		
TFP	-0.0889 (0.16)	-0.0738 (0.15)	-0.0084 (1.45E-2)	-0.0069 (1.42E-2)
Idiosyncratic Demand Factors	1.1047*** (0.36)	1.0115*** (0.24)	0.1038*** (0.03)	0.0950*** (0.02)
Capital	-3.0531E-6 (9.05E-6)		2.8683E-7 (8.23E-7)	
Year dummies	included	included		

Notes: Standard errors are in parentheses. Marginal effects are effects of a 1 unit increase in the corresponding independent variable. *** means significant at 1% level. ** means significant at 5% level.

Table 8: Effects of TFP and Idiosyncratic Demand Factors on Plant Exit and Ownership Change: Multinomial Logit Parameter Estimates

Variables	Exit		Ownership change	
	(1)	(2)	(1)	(2)
Constant	-2.1272*** (0.61)	-2.0353*** (0.52)	-2.6482*** (0.68)	-2.7072*** (0.60)
TFP	0.1958 (0.15)	0.1901 (0.15)	-0.0662 (0.16)	-0.0529 (0.16)
Idiosyncratic Demand Factors	-0.7721*** (0.21)	-0.7217*** (0.18)	1.0096*** (0.37)	0.9275*** (0.25)
Capital	3.4449E-6 (7.82E-6)		2.5128E-6 (9.16E-6)	
Year dummies	included	included		

Notes: Standard errors are in parentheses. *** means significant at 1% level. ** means significant at 5% level. * means significant at 10% level.

Table 9: Marginal Effects of TFP and Idiosyncratic Demand Factors on Probabilities of Plant Survival, Exit and Ownership Change: Multinomial Logit

Variables Groups	Productivity		Demand-specific Factor		Capital
	(1)	(2)	(1)	(2)	(1)
Exit	0.0205 (1.54E-2)	0.0199 (1.54E-2)	-0.0881** (2.00E-2)	-0.0823*** (1.67E-2)	3.2595E-7 (7.58E-7)
Survival	-0.0124 (1.97E-2)	-0.0131 (1.93E-2)	-0.0142 (3.83E-2)	-0.0117 (2.78E-2)	-5.2919E-7 (1.08E-6)
Ownership Change	-0.0081 (1.46E-2)	-0.0068 (1.43E-2)	0.1023*** (3.46E-2)	0.0941*** (2.39E-2)	2.0324E-7 (8.26E-7)

Notes: Standard errors are in parentheses. Marginal effects are effects of a 1 unit increase in the corresponding independent variable. ** means significant at 5% level.

Table 10: Summary Statistics for Firm-Level TFP and Idiosyncratic Demand Factors

Type	N	TFP		Idiosyncratic Demand Factors	
		Mean	Std. Dev.	Mean	Std. Dev.
Expanding Firms	27	2.3977	0.46	-0.5529	1.14
Other Firms	138	2.5667	1.10	-1.4384	1.48
Total	165	2.5389	1.03	-1.2926	1.46

Table 11: Firm-level Demand Estimation Results using 2SLS

Variables	Estimates	Standard Error	P-value
Constant	12.9076***	0.2315	<0.0001
Log Price	-1.2157***	0.4977	0.0157
Single-Plant dummy	-2.2082***	0.1592	<0.0001
Tests		F-value	P-value
Basmann's Overidentification Test		2.61	0.1084
First-stage F Test for Weak IVs		59.55	<0.0001

Note: *** means significant at 1% level.

Table 12: Effects of TFP and Idiosyncratic Demand Factors on Firm Expansion: Logit Model

Variables	Parameter Estimates		Marginal Effects	
	(1)	(2)	(1)	(2)
Constant	-1.8222 (5.17)	-2.7427 (4.01)		
TFP	0.0689 (1.17)	0.0999 (1.00)	0.0036 (0.04)	0.0052 (0.04)
Idiosyncratic Demand Factors	1.5600 (1.48)	0.9803 (0.92)	0.0807*** (0.03)	0.0511** (0.02)
Capital	-2.7795E-6 (5.20E-6)		-1.4325E-7 (1.73E-7)	
Year dummies	included	included		

Notes: Standard errors are in parentheses. Marginal effects are effects of a 1 unit increase in the corresponding independent variable. * means significant at 10% level. ** means significant at 5% level. *** means significant at 1% level.